Fleet Allocation: one exact MIP

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1. **Forecast** base revenue and demand from 8 history weeks
2. **One MIP** that reproduces the scorer's rules exactly, written as an SAA over demand scenarios
3. **Our own scorer** on 500 fresh scenarios, to pick the plan without overfitting to the public scenario

Level 3 plan, valid: **4.11 M** expected profit vs. 3.12 M for the baseline (+32 %)

Forecast

- **Base revenue:** ratio estimator per station × category, error $< 2\,\%$, independent of demand $\Rightarrow$ a point estimate is enough
- **Rental length:** one geometric distribution (mean ≈ 3 days), independent of lead time and weekday
- **Demand is not Poisson:** the overdispersion persists within a week and resets on Monday

$$P_{skd} \mid G \sim \mathrm{Poi}\big(G_{skw}\,\lambda_{sk,\mathrm{wd}}\big), \qquad G \sim \Gamma\big(\tfrac{1}{v}, \tfrac{1}{v}\big), \quad \hat{v} = 0.017\ (\pm 13\,\%)$$

- **Bookings** update G (conjugate Bayes): forecast error $-28\,\%$ in the backtest

Model: SAA on a time-space network

- **Plan (first stage):** add or remove cars, transfer them by driver or truck (8 places); fleet flow with lead times; fleet $\geq$ committed car-days
- **Per scenario (second stage):** serving, one-step upgrades, fare buckets with marginal revenues

$$\max \ \frac{1}{|\Omega|} \sum_{\omega} \text{revenue}^{\omega} \ - \ \text{holding} \ - \ \text{addition} \ - \ \text{transfer cost}$$

- **Point forecast** = SAA with one scenario, $D = E[D]$
- **Exact "own category first" rule** (one binary per cell and scenario): **+100 k**. Without it, the model plans upgrades the scorer never makes.

Results (our scorer, 500 test scenarios)

Plan	Note	Profit
Point forecast (submitted)	gap 0.6 %	**4.11 M**
SAA, 20 scenarios	gap 3.7 %	4.10 M
Revenue curves	$E[\mathrm{rev}]$ as a function of capacity	4.06 M
Point, without exact rule		3.99 M
No transfers (Level 1)		3.42 M
Baseline	fleetkit rule	3.12 M

- Final plan: 233 cars removed, 898 transferred (mostly by truck)
- Transfers ($+590$ k) $\gg$ exact rules ($+100$ k) $\gg$ choice of revenue model

Limits and next steps

Limits

- Only 8 weeks of history: no train/test split; the scorer uses our own demand model
- SAA not converged in the time available (3–4 % gap)
- Solved with Gurobi; the model is plain PuLP and also runs with HiGHS

Next steps

- Rolling re-planning as bookings arrive
- Decomposition for larger SAA
- For SIXT: weekly fleet balancing and defleeting